



Barker College

Student No:

2012
TRIAL
HIGHER SCHOOL
CERTIFICATE

Mathematics Extension 1

ANSWER SHEET

Section I – Multiple Choice

Choose the best response and fill in the response oval completely.

**Start
Here** →

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

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Mathematics Extension 1

Staff Involved:

Friday 10th August

- RMH* • GIC
- GPF* • PJR
- BJB • BHC
- VAB

Number of copies: 90

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 - 14.

Total marks – 70

Section I Pages 2 - 5

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II Pages 6 - 10

60 marks

- Attempt Questions 11 - 14
- Allow about 1 hours 45 minutes for this section

Section I — Multiple Choice

Attempt Questions 1 - 10

Use the multiple-choice answer sheet for Questions 1 – 10.

1. The x -coordinate of the point which divides the interval joining $A(3,1)$ and $B(-1, 5)$ externally in the ratio 4 : 3 is:

(A) -13 (B) $\frac{5}{7}$ (C) 17 (D) 15

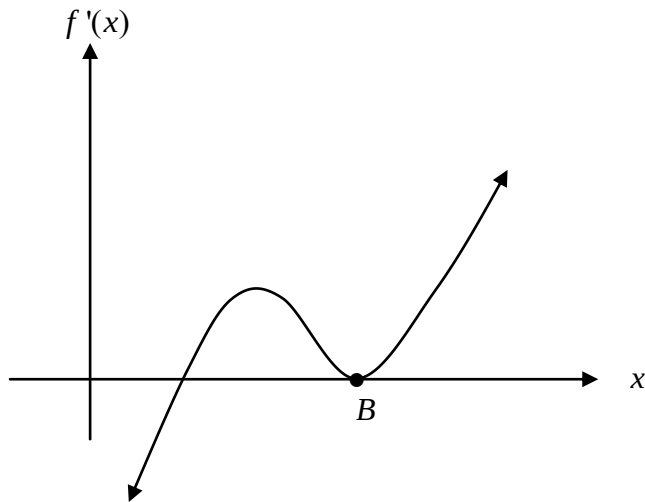
2. $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$ is equal to :

(A) 0 (B) 1 (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

3. $3(\sin^{-1} x + \cos^{-1} x)$ is equal to :

(A) 1 (B) $\frac{3\pi}{2}$ (C) 3 (D) 3π

4.



NOT TO SCALE

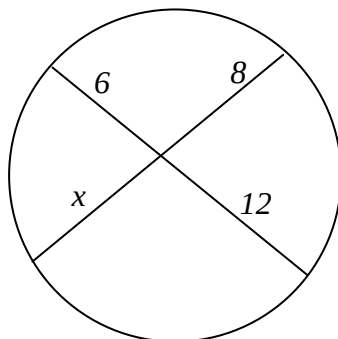
The gradient function of $y = f(x)$ is shown above. On the curve $y = f(x)$, B would be a :

- (A) maximum turning point
- (B) point of inflexion
- (C) horizontal point of inflexion
- (D) minimum turning point

5. Given that $f^{-1}(x) = \frac{2x}{x-1}$, $f(x)$ would have the equation :

- (A) $\frac{x}{x+2}$
- (B) $\frac{y}{y-2}$
- (C) $\frac{2y}{y-1}$
- (D) $\frac{x}{x-2}$

6.



NOT TO SCALE

The value of x is:

- (A) 9
- (B) 4
- (C) 5.5
- (D) 1

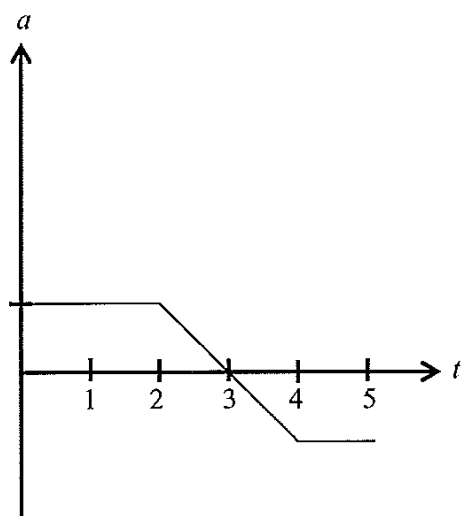
7. A particle moves in simple harmonic motion such that $v^2 = -4(x-5)(x+1)$,

where v is velocity in m/s.

Maximum acceleration of this particle happens when:

- (A) $x = 5$ and -1 (B) $x = 2$ (C) $x = -5$ and 1 (D) $x = 0$

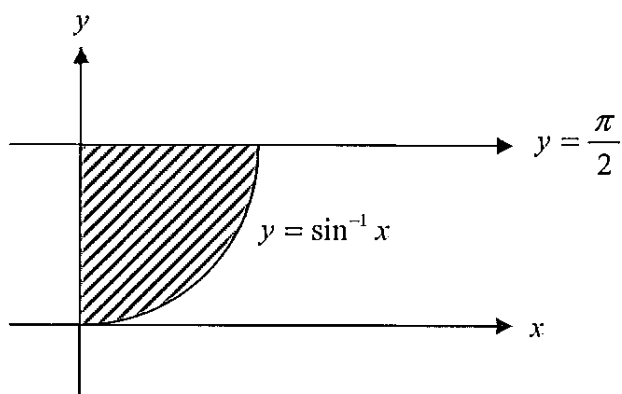
8. The acceleration-time graph of a particle is shown below.



The time(s) when the particle has a maximum velocity is :

- (A) $t = 2$ (B) $t = 3$ (C) $t = 4$ (D) $0 < t < 2$

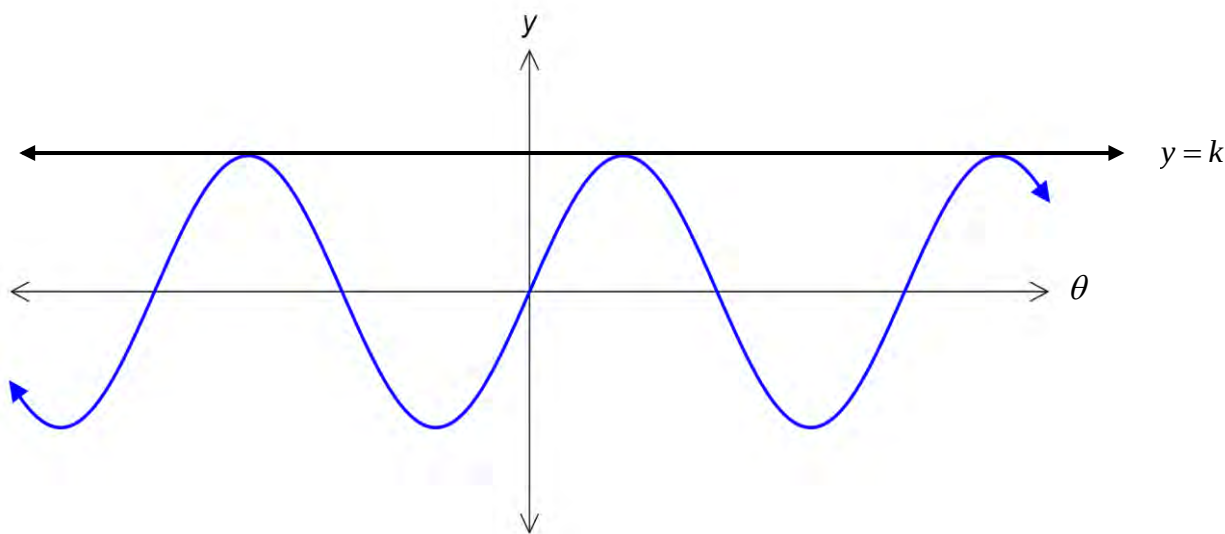
9. The area bounded by the y -axis, $y = \frac{\pi}{2}$ and $y = \sin^{-1} x$ is shaded in the diagram below.



The value of this area, in square units is:

- (A) $\frac{\pi}{2} - 1$ (B) 1 (C) $\frac{\pi}{2}$ (D) $1 - \frac{\pi}{2}$

10. Part of the curve $y = 2 \sin 2\theta$ is drawn below.



The horizontal line $y = k$ is also drawn as shown above.

The general solution of the intersection of these two functions is:

- (A) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$ (B) $2n\pi \pm \frac{\pi}{4}$ (C) $n\pi + (-1)^n \frac{\pi}{2}$ (D) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{2}$

End of Section I

(a) Solve $\frac{x}{3-2x} < 1$. 4

(b) Evaluate $\int_{-1}^1 \frac{dx}{\sqrt{4-x^2}}$, leaving your answer in exact form. 3

(c) Evaluate $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \cos^2 2x \, dx$, leaving your answer in exact form. 4

(d) Using the substitution $x = u - 2$, or otherwise, find $\int \frac{x+1}{\sqrt{(x+2)^3}} \, dx$. 4

End of Question 11

- (a) Prove by induction that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1)$$

for all $n \geq 1$, where n is an integer.

3

- (b) Find the coefficient of x^2 in the expansion of $\left(\frac{x^4}{2} + \frac{2}{x}\right)^8$.

3

- (c) Consider the function $f(x) = 3\sin^{-1}x$.

- (i) State the domain and range of the function.

2

- (ii) Draw a neat sketch of the function.
Clearly label all essential features.

1

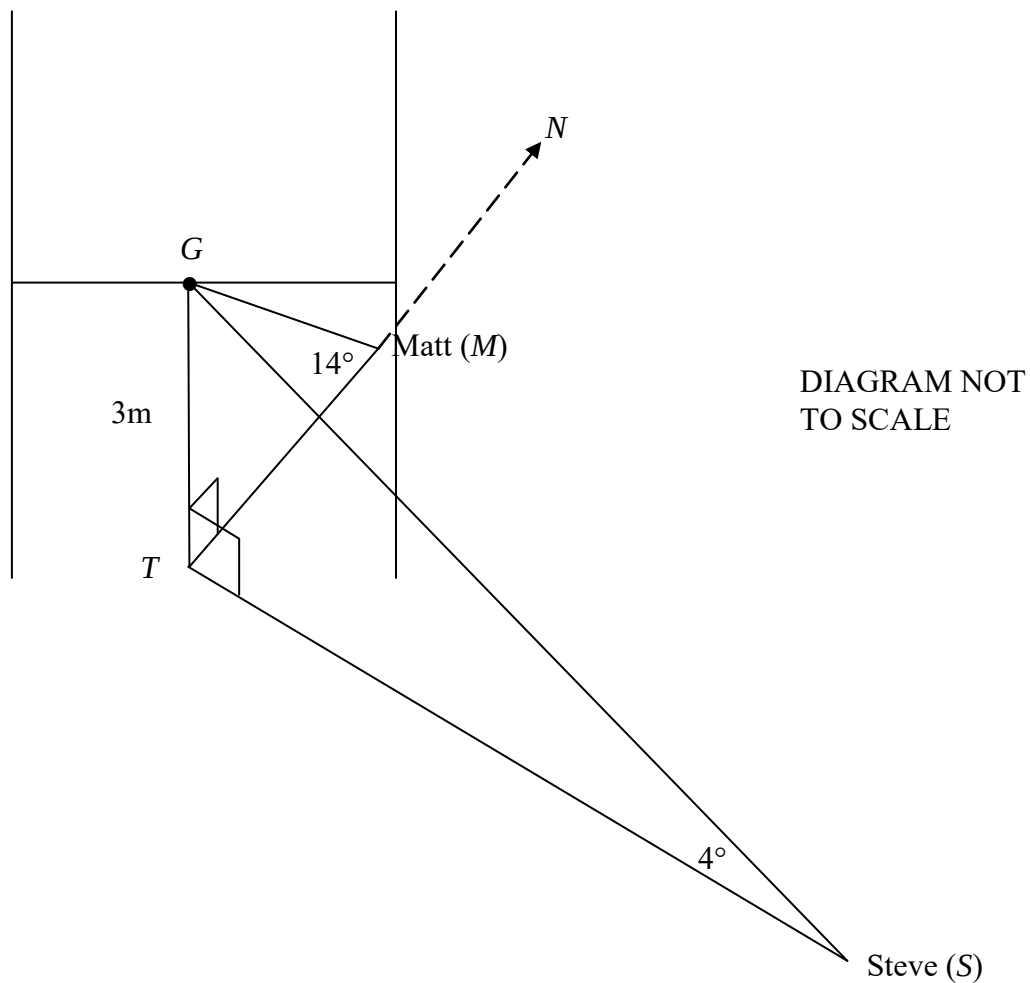
- (iii) Find the gradient of the tangent to the function $f(x) = 3\sin^{-1}x$

2

at the point $\left(\frac{1}{2}, \frac{\pi}{2}\right)$.

Question 12 continues on page 8

(d)



Matt (*M*) and Steve (*S*) operate ground-level cameras at a rugby match. The cameras are aimed at the centre of the cross-bar of the goal post (*G*) which is 3 metres above the ground.

Matt is directly north of the centre of the goal post.

The centre of the goal post is on a bearing of 302° from Steve.

The cameras operated by Matt and Steve are set at angles of elevation of 14° and 4° respectively.

(i) Show that $\angle MTS = 122^\circ$. 1

(ii) Calculate the distance between the two cameras, to the nearest centimetre. 3

End of Question 12

- (a) (i) Express $8 \cos \theta - 6 \sin \theta$ in the form $R \cos(\theta + \alpha)$ where α is in degrees. 2
- (ii) Hence, or otherwise, find the solutions of the equation $8 \cos \theta - 6 \sin \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$. 3
- (b) A particle moves in such a way that its displacement, x cm, from the origin at any time is given by the function $x = 2 + \cos^2 t$ where t is in seconds.
- (i) Show that acceleration is given by $\ddot{x} = 10 - 4x$. 2
- (ii) Find the centre of the motion. 1
- (iii) Find the amplitude of the motion. 1
- (c) A ball is kicked on level ground to just clear a fence 2 m high and 40 m away. The initial velocity is 30 m/s and the angle of projection is α .
- The displacement equations are
- $$x = 30t \cos \alpha \quad \text{and} \quad y = -5t^2 + 30t \sin \alpha \quad (\text{DO NOT PROVE THESE}).$$
- (i) Show that $y = \frac{-x^2}{180} \sec^2 \alpha + x \tan \alpha$. 2
- (ii) Hence, or otherwise, find the angles of projection that allow the ball to just clear the fence. Answer to the nearest degree. 4

End of Question 13

Question 14 (15 marks)**[START A NEW BOOKLET]****Marks**

- (a) $P(2p, p^2)$ and $Q(2q, q^2)$ are two points on the parabola $x^2 = 4y$.
The chord PQ subtends a right angle at the origin.
- (i) Show that $pq = -4$. **1**
- (ii) If M is the midpoint of PQ , find the locus of M . **2**
- (b) Consider the function $f(x) = \frac{x+2}{x^2+1}$.
- (i) Find the points where the curve crosses the x -axis and y -axis. **1**
- (ii) Find the x -coordinates of any stationary points, and without finding the second derivative, determine their nature. **3**
- (iii) Describe the behaviour of $y = f(x)$ as $x \rightarrow \pm \infty$. **1**
- (iv) Sketch the curve $y = f(x)$ using an appropriate scale and showing all the information above. Label the axes and any critical points. **1**
- (c) The acceleration of a particle moving along a straight path is given by
- $$\ddot{x} = -\frac{e^x + 1}{e^{2x}}, \text{ where } x \text{ is in metres.}$$
- Initially the particle is at the origin with a velocity of 2 m/s.
- (i) show that $v = e^{-x} + 1$. **3**
- (ii) Find the equation of the displacement, x , in terms of t . **3**

End of Question 14
End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

2012 yr 12 Extension 1 Trial HSC

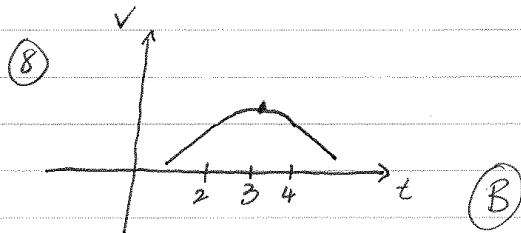
Multiple Choice

$$\begin{aligned} (1) & \left(\frac{x_2 k + x_1 l}{k+l}, \right) \\ &= \left(\frac{-1 \times 4 + 3 \times 3}{-4+3}, \right) \\ &= \left(\frac{4+9}{-1}, \right) \\ &= (-13, 0) \quad (A) \end{aligned}$$

$$\begin{aligned} (6) \quad 8x &= 72 \\ x &= 9 \quad (A) \end{aligned}$$

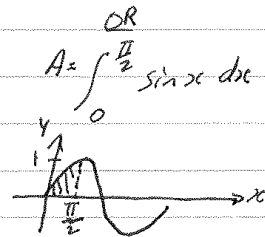
$$\begin{aligned} (7) \quad \text{Max acceleration @ } v=0 \\ \therefore x=5, -1 \quad (A) \end{aligned}$$

$$\begin{aligned} (2) \quad \lim_{x \rightarrow 0} \frac{3}{2} x \frac{\tan 3x}{3x} \\ = \frac{3}{2} \quad (C) \end{aligned}$$



$$\begin{aligned} (3) \quad (\sin^{-1} x + \cos^{-1} x) &= \frac{\pi}{2} \\ \therefore \frac{3\pi}{2} \quad (B) \end{aligned}$$

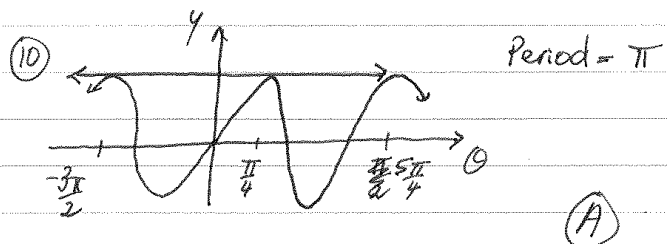
$$\begin{aligned} (9) \quad x &= \sin y \\ \int_0^{\pi/2} \sin y \, dy \\ &= [-\cos y]_0^{\pi/2} \\ &= -\cos \frac{\pi}{2} + \cos 0 \end{aligned}$$



$$\begin{aligned} (4) \quad \begin{array}{c} + & 0 & + \\ / & - & / \end{array} \\ (C) \end{aligned}$$

$$\begin{aligned} &= 0 + 1 \\ &= 1 \quad (B) \end{aligned}$$

$$\begin{aligned} (5) \quad x &= \frac{2y}{y-1} \\ xy - x &= 2y \\ xy - 2y &= x \\ y(x-2) &= x \\ y &= \frac{x}{x-2} \quad (D) \end{aligned}$$



n	0	1	2
$\frac{1\pi + (-1)^n \pi}{4}$	$\frac{\pi}{4}$	$\frac{\pi}{4}$	$\frac{5\pi}{4}$

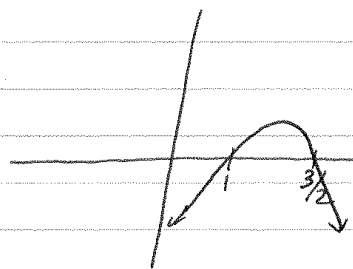
$$\text{Q4 1(a)} \quad \frac{(3-2x)^2 x}{(3-2x)} < 1(3-2x)^2$$

$$x(3-2x) < (3-2x)^2$$

$$x(3-2x) - (3-2x)^2 < 0$$

$$(3-2x)[x - (3-2x)] < 0$$

$$(3-2x)(3x-3) < 0$$



$$x < 1, \quad x > 3/2$$

$$\begin{aligned} \text{(b)} \quad \left[\sin^{-1} \frac{x}{2} \right]_{-1}^1 &= \frac{\pi}{6} - -\frac{\pi}{6} \\ &= \pi/3 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \cos 2x &= 2\cos^2 x - 1 \\ \cos^2 x &= \frac{1}{2}(1 + \cos 2x) \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \int_{\pi/12}^{\pi/4} (1 + \cos 4x) dx &= \frac{1}{2} \left[x + \frac{1}{4} \sin 4x \right]_{\pi/12}^{\pi/4} \\ &= \frac{1}{2} \left[\left(\frac{\pi}{4} + \frac{1}{4} \times 0 \right) - \left(\frac{\pi}{12} + \frac{1}{4} \sin \frac{\pi}{3} \right) \right] \\ &= \frac{1}{2} \left(\frac{\pi}{4} - \frac{\pi}{12} - \frac{\sqrt{3}}{8} \right) \\ &= \frac{1}{2} \left(\frac{\pi}{4} - \frac{\pi\sqrt{3}}{24} \right) \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad u &= x+2 \\ \frac{du}{dx} &= 1 \\ du &= dx \end{aligned} \quad \int \frac{u-1}{u^{3/2}} du$$

$$= \int (u^{-1/2} - u^{-3/2}) du$$

$$\begin{aligned} &= \frac{u^{1/2}}{1/2} - \frac{u^{-1/2}}{-1/2} + C \\ &= 2\sqrt{u} + \frac{2}{\sqrt{u}} + C = 2\sqrt{x+2} + \frac{2}{\sqrt{x+2}} + C \end{aligned}$$

Qn 12

(a) Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= 1^2 \\ &= 1 \\ \text{RHS} &= \frac{1}{6} \times 1 \times (1+1) \times (2+1) \\ &= \frac{1}{6} \times 1 \times 2 \times 3 \\ &= 1 \end{aligned}$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$1^2 + 2^2 + \dots + k^2 = \frac{1}{6} k(k+1)(2k+1)$$

Prove true for $n=k+1$

$$\begin{aligned} \text{LHS} &= 1^2 + 2^2 + \dots + k^2 + (k+1)^2 \\ \text{RHS} &= \frac{1}{6} k(k+1)(2k+1) + (k+1)^2 \\ &= \frac{k+1}{6} [k(2k+1) + 6(k+1)] \\ &= \frac{k+1}{6} (2k^2 + k + 6k + 6) \\ &= \frac{k+1}{6} (2k^2 + 7k + 6) \\ &= \frac{k+1}{6} (2k+3)(k+2) \\ &= \frac{1}{6} (k+1)((k+1)+1)(2(k+1)+1) \\ &= S_{k+1} \end{aligned}$$

If statement true for $n=k$, then true for $n=k+1$.

Since true for $n=1$, then true for $n=2$

& as true for $n=2$, then true for $n=3$ & so on.

$\therefore$ True for all $n \geq 1$ (where n is an integer)

Qn 12 (Continued)

$$(b) {}^8C_r \left(\frac{x^4}{2}\right)^{8-r} \left(\frac{2}{x}\right)^r$$

$$= {}^8C_r \left(\frac{1}{2}\right)^{8-r} \cdot 2^r \cdot x^{32-5r}$$

$$32-5r=2$$

$$30=5r$$

$$\therefore r=6$$

$\therefore$ Coefficient

$$= {}^8C_6 \times \left(\frac{1}{2}\right)^2 \times 2^6$$

$$= 448$$

OR

$$(1+x)^8 = 1 + 8x + 28x^2 + 56x^3 + 70x^4 + 56x^5 + \boxed{28x^6} + 8x^7 + 1x^8$$

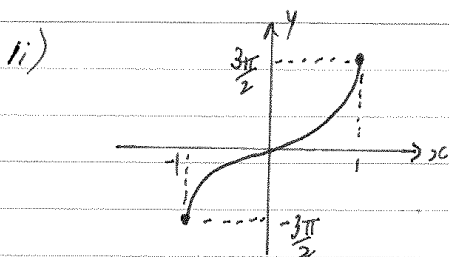
$$\left(\frac{x^4 + 2}{2}\right)^8 = \left(\frac{x^4}{2}\right)^8 + 8\left(\frac{x^4}{2}\right)^7 \left(\frac{2}{x}\right) + 28\left(\frac{x^4}{2}\right)^6 \left(\frac{2}{x}\right)^2 + \dots$$

$$\begin{array}{ccc} x^{32} & x^{27} & x^{22} \\ \xrightarrow{-5} & \xrightarrow{-5} & \xrightarrow{-5} \end{array}$$

$$28\left(\frac{x^4}{2}\right)^2 \left(\frac{2}{x}\right)^6 = \frac{28 \times 64}{4} x^2 \quad \therefore \underline{\underline{448}}$$

(c) i) D: $-1 \leq x \leq 1$

R: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$



Qn 12 (Continued)

$$\text{iii)} \quad f'(x) = \frac{3}{\sqrt{1-x^2}}$$

$$f'\left(\frac{1}{2}\right) = \frac{3}{\sqrt{1-\frac{1}{4}}}$$

$$= \frac{3 \div \frac{\sqrt{3}}{2}}{\frac{2}{2}}$$

$$= \frac{6}{\sqrt{3}}$$

$$= 2\sqrt{3}$$

Eqn of tangent @ $\left(\frac{1}{2}, \frac{\pi}{2}\right)$

$$y - \frac{\pi}{2} = \frac{6}{\sqrt{3}}\left(x - \frac{1}{2}\right)$$

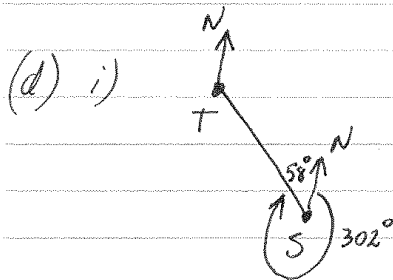
$$\sqrt{3}y - \frac{\pi\sqrt{3}}{2} = 6x - 3$$

OR

$$y - \frac{\pi}{2} = 2\sqrt{3}\left(x - \frac{1}{2}\right)$$

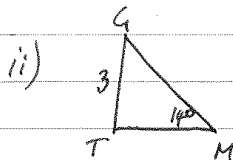
$$y = 2\sqrt{3}x - \sqrt{3} + \frac{\pi}{2}$$

$$y = 2\sqrt{3}x - \left(\sqrt{3} - \frac{\pi}{2}\right)$$

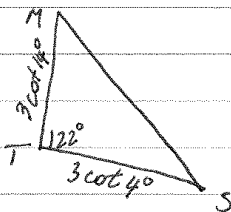


$$\angle MTS = 180 - 58 \quad (\text{interior } \angle \text{ in } \parallel \text{ lines})$$

$$= 122^\circ$$



$$MT = 3 \cot 14^\circ$$



$$x^2 = (3 \cot 14^\circ)^2 + (3 \cot 40^\circ)^2 - 2(3 \cot 14^\circ)(3 \cot 40^\circ) \cos 122^\circ$$

$$\hat{=} 2532.46$$

$$x \hat{=} 50.32 \text{ m}$$

Qu 13

$$\begin{aligned} \text{a) i) } R \cos(\theta + \alpha) &= R [\cos \theta \cos \alpha - \sin \theta \sin \alpha] \\ &= 10 \left(\cos \theta \left(\frac{8}{10} \right) - \frac{6}{10} \sin \theta \right) \end{aligned}$$



$$\therefore \cos \alpha = \frac{8}{10}$$

$$\alpha = 36^\circ 52' \quad \therefore 10 \cos(\theta + 36^\circ 52')$$

$$\text{ii) } 10 \cos(\theta + 36^\circ 52') = 5$$

$$\cos(\theta + 36^\circ 52') = 0.5$$

$$\theta + 36^\circ 52' = 60^\circ \text{ or } 300^\circ$$

$$\therefore \theta = 23^\circ 08' \text{ or } 263^\circ 08'$$

$$\text{b) i) } x = 2 + (\cos t)^2 \quad \cos^2 t = x - 2$$

$$\frac{dx}{dt} = -2 \sin t \cos t$$

$$= -\sin 2t$$

$$\frac{d^2x}{dt^2} = -2 \cos 2t$$

$$= -2(2 \cos^2 t - 1)$$

$$= -4(x - 2) + 2$$

$$= -4x + 8 + 2$$

$$= 10 - 4x$$

$$\text{OR } \frac{dx}{dt} = -2 \sin t \cos t$$

$$\frac{d^2x}{dt^2} = -2 \sin t (-\sin t) + (-2 \cos t) \cos t$$

$$= 2 \sin^2 t - 2 \cos^2 t$$

$$= 2(\sin^2 t - \cos^2 t)$$

$$= 2(1 - \cos^2 t - \cos^2 t)$$

$$= 2(1 - 2 \cos^2 t)$$

$$= 2 - 4 \cos^2 t$$

$$= 2 - 4(x - 2)$$

$$= 2 - 4x + 8$$

$$= 10 - 4x$$

$$\text{ii) } a = 0$$

$$0 = 10 - 4x$$

$$\therefore x = 2.5 \text{ (centre of motion)}$$

$$\text{iii) } 0 \leq \cos^2 t \leq 1$$

$$\therefore 2 \leq x \leq 3$$

$$\text{amplitude} = \frac{1}{2} \text{ cm}$$

$$v = 0$$

$$0 = -\sin 2t$$

$$\text{OR } 2t = 0, \pi, 2\pi, \dots$$

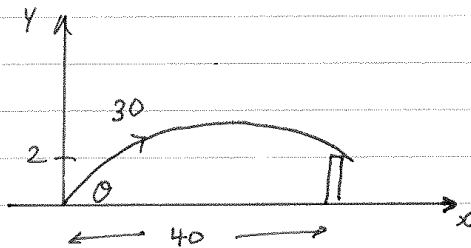
$$t = 0, \frac{\pi}{2}, \pi, \dots$$

$$t = 0, x = 3 \text{ cm} \quad \therefore \text{amplitude}$$

$$t = \frac{\pi}{2}, x = 2 \text{ cm} \quad = \frac{1}{2} \text{ cm}$$

Qn 13 Continued

(c)



$$i) \quad t = \frac{x}{30 \cos \alpha} \quad \therefore y = \frac{-5x^2}{900 \cos^2 \alpha} + \frac{30x \sin \alpha}{30 \cos \alpha}$$

$$y = \frac{-x^2 \sec^2 \alpha}{180} + x \tan \alpha$$

$$ii) \quad 2 = \frac{-40^2 \sec^2 \alpha}{180} + 40 \tan \alpha$$

$$360 = -40^2 \sec^2 \alpha + 7200 \tan \alpha$$

$$360 = -40^2 (\tan^2 \alpha + 1) + 7200 \tan \alpha$$

$$-9 = 40 (\tan^2 \alpha + 1) - 180 \tan \alpha$$

$$0 = 40 \tan^2 \alpha - 180 \tan \alpha + 49$$

$$\tan \alpha = \frac{180 \pm \sqrt{180^2 - 4(40)(49)}}{2(40)}$$

$$= \frac{180 \pm \sqrt{24560}}{80}$$

$$\therefore \alpha = 76^\circ 38', 16^\circ 14' \quad 16^\circ 14' < \alpha < 76^\circ 38'$$

$$\therefore \alpha = 16^\circ, 77^\circ$$

Qu 14

a) i) $m_{OP} \times m_{OQ} = -1$

$$\frac{p^2-0}{2p-0} \times \frac{q^2-0}{2q-0} = -1$$

$$\frac{p \times q}{2 \times 2} = -1$$

$$\boxed{pq = -4}$$

ii) $M = \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$
 $= \left(p+q, \frac{p^2+q^2}{2} \right)$

$$y = \frac{1}{2} \left((p+q)^2 - 2pq \right)$$

$$= \frac{1}{2} (x^2 - 2x - 4)$$

$$y = \frac{1}{2} x^2 + 4$$

b) i) $0 = \frac{x+2}{x^2+1}$ (x-intercept)

$$\therefore x = -2 \quad (-2, 0)$$

$$y = \frac{0+2}{0+1}$$
 (y-intercept)

$$= 2 \quad (0, 2)$$

Question 14 Continued

$$\begin{aligned} \text{ii) } f'(x) &= \frac{(x^2+1)(1) - (x+2)(2x)}{(x^2+1)^2} \\ &= \frac{-x^2 - 4x + 1}{(x^2+1)^2} \end{aligned}$$

$$f'(x) = 0 \quad \text{when} \quad -x^2 - 4x + 1 = 0$$

$$0 = x^2 + 4x - 1$$

$$x = \frac{-4 \pm \sqrt{16 - 4(1)(-1)}}{2}$$

$$x = \frac{-4 \pm \sqrt{20}}{2}$$

$$= -2 \pm \sqrt{5}$$

x	-5	$-2 - \sqrt{5}$	0	$-2 + \sqrt{5}$	1
$f'(x)$	$-4/26^2$	0	1	0	$-4/4$

$\diagdown$ --- $\diagup$ --- $\diagdown$

Min

Max

$$\text{iii) } \lim_{x \rightarrow \infty} \left(\frac{x+2}{x^2+1} \right) = \lim_{x \rightarrow \infty} \left(\frac{\frac{x}{x^2} + \frac{2}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}} \right)$$

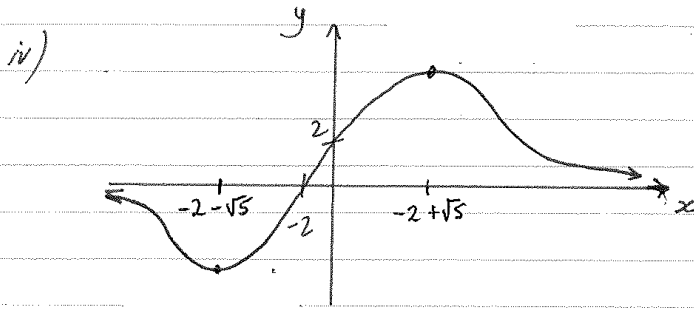
$$= \frac{0}{1}$$

$$= 0^+$$

$$x \rightarrow \infty, y \rightarrow 0^+$$

$$x \rightarrow -\infty, y \rightarrow 0^-$$

Question 14 Continued



(c) i) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$\frac{1}{2} v^2 = \int (-e^{-x} - e^{-2x}) dx$$

$$= e^{-x} + \frac{1}{2} e^{-2x} + C$$

When $x=0, v=2$

$$\therefore \frac{2^2}{2} = e^0 + \frac{1}{2} e^0 + C$$

$$C = \frac{1}{2}$$

$$\frac{1}{2} v^2 = e^{-x} + \frac{1}{2} e^{-2x} + \frac{1}{2}$$

$$v^2 = 2e^{-x} + e^{-2x} + 1$$

$$= e^{-2x} + 2e^{-x} + 1$$

$$= (e^{-x} + 1)^2$$

$$\therefore v = \pm (e^{-x} + 1)$$

When $x=0, v=2$

$$\therefore v = e^{-x} + 1$$

Qn 14 Continued

ii)

$$\frac{dx}{dt} = e^{-x} + 1$$

$$\frac{dt}{dx} = \frac{1}{e^{-x} + 1}$$

$$t = \int \frac{1}{e^{-x} + 1} dx$$

$$= \int \frac{e^x}{e^x} \cdot \frac{1}{e^{-x} + 1} dx$$

$$= \int \frac{e^x}{e^0 + e^x} dx$$

$$= \log_e (1 + e^x) + C$$

When $t=0$, $x=0$

$$0 = \ln(e^0 + 1) + C$$

$$\therefore C = -\ln 2$$

$$\begin{aligned}\therefore t &= \ln(e^x + 1) - \ln 2 \\ &= \ln\left(\frac{e^x + 1}{2}\right)\end{aligned}$$

$$e^t = \frac{e^x + 1}{2}$$

$$2e^t = e^x + 1$$

$$e^x = 2e^t - 1$$

$$x = \ln(2e^t - 1)$$